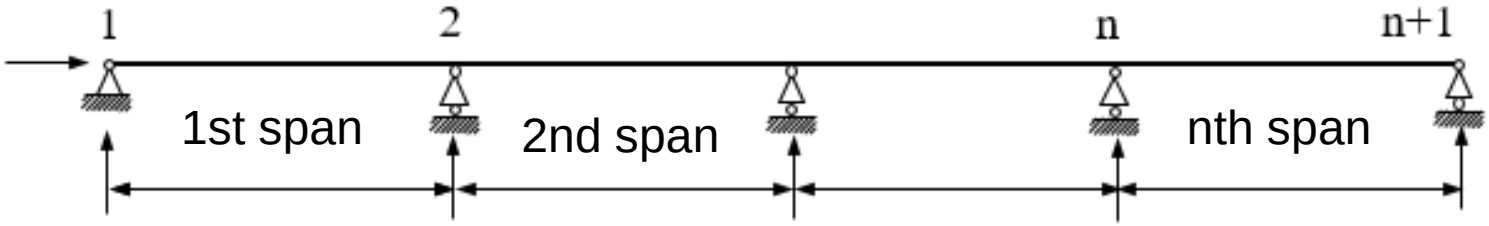
STRUCTURAL ANALYSIS 1 (INS2512) - LESSON 6 -

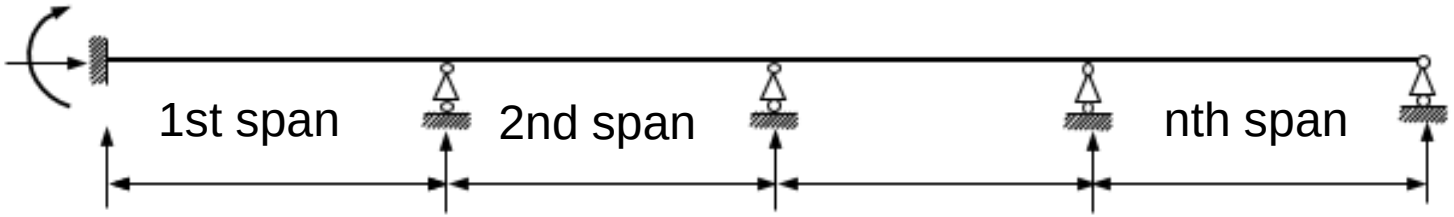
Assist. Prof. Dr. Muzaffer BÖREKÇİ
Email: structuralanalysis1mb@gmail.com

Gerber Beams

A beam having more than one span and resting on the supports, one of which is either a fixed support or a pin support and the rest are roller supports, is referred to as a continuous beam.

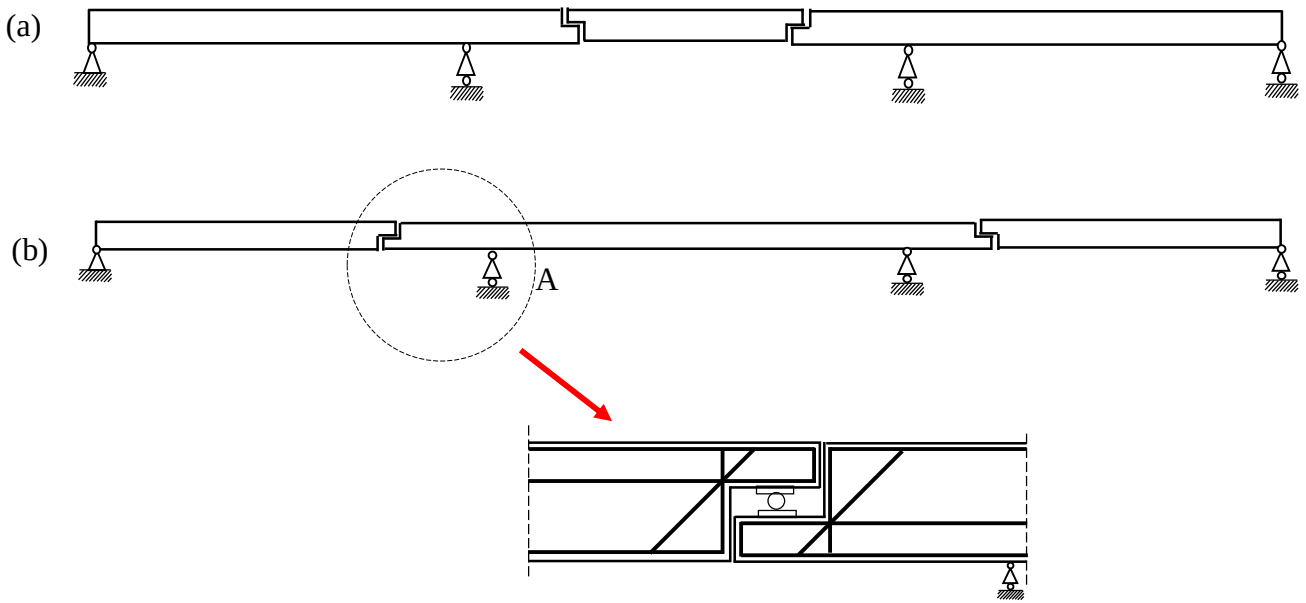


Number of support reactions = n+2

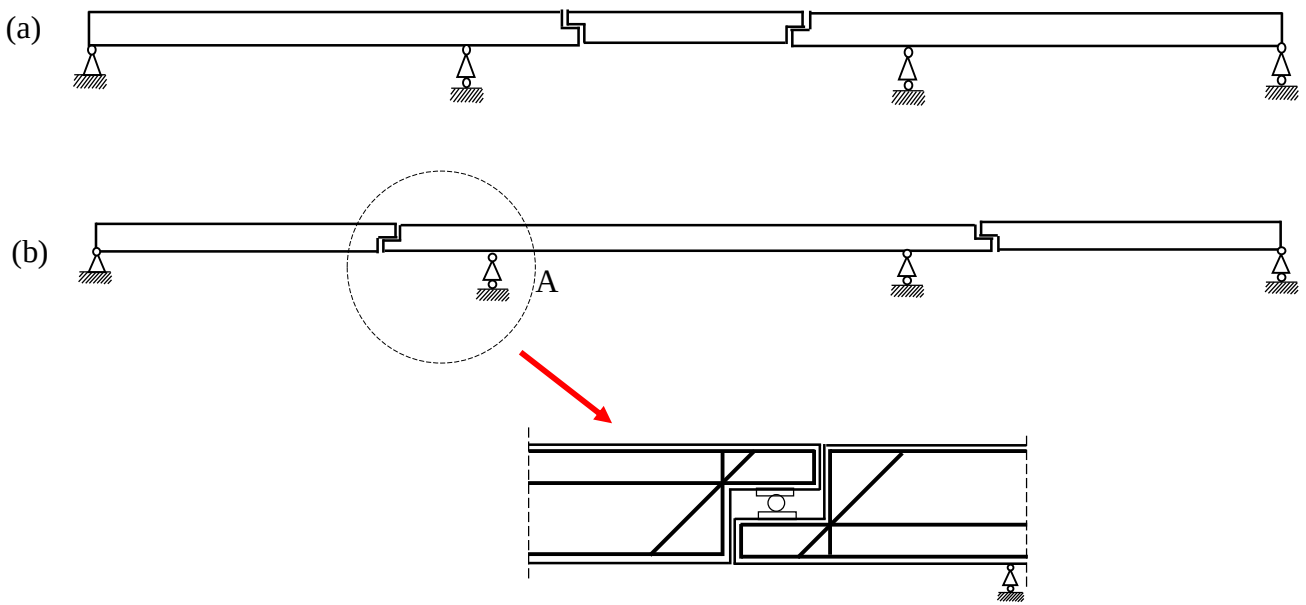


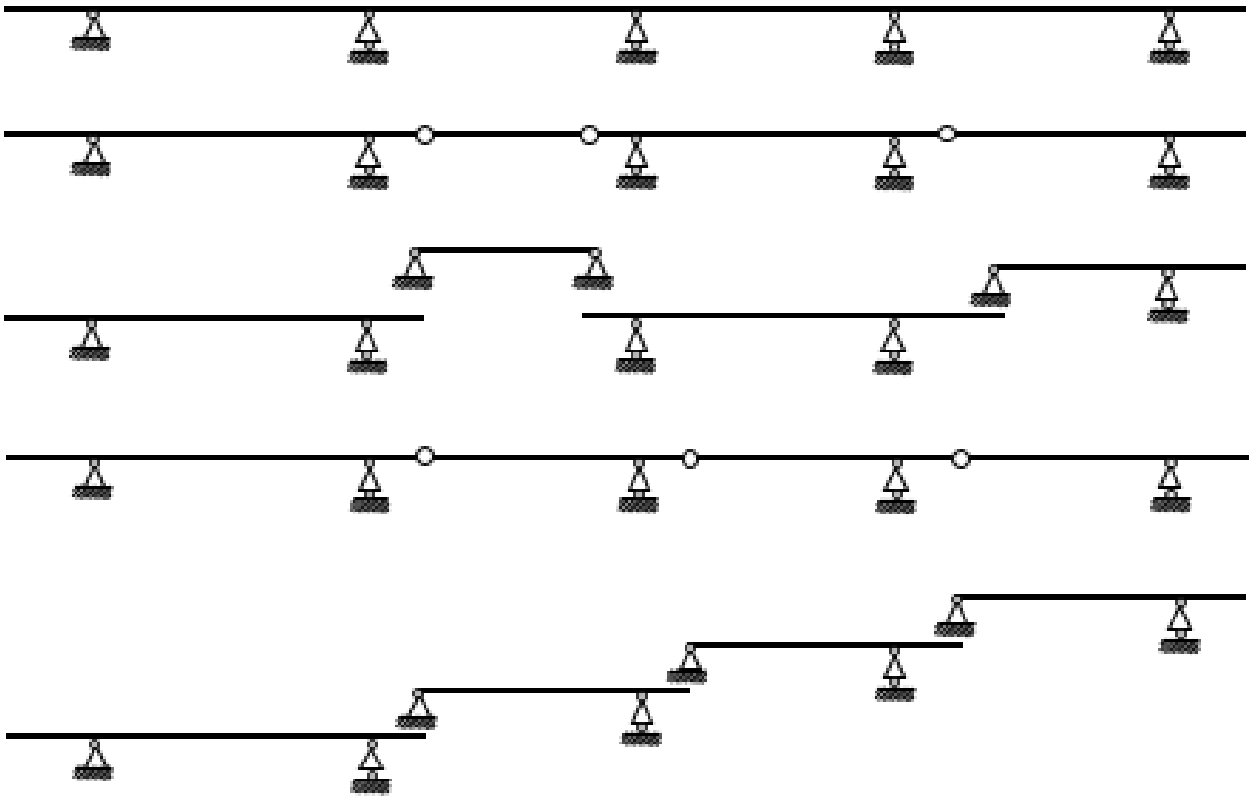
Number of support reactions = n+3

A typical example of continuous beams is multi-span bridges. Any settlement in the piers (especially in middle piers) will cause shear force and bending moment in the beam. In order to prevent internal forces caused by support settlements, statically indeterminate continuous beams are transformed into statically determinate structures by introducing hinges at proper locations.



The resulting statically determinate beams are referred to as Gerber beams.





In order to make the continuous beam a Gerber beam, as many hinges as the degree of indeterminacy should be introduced.

While introducing the hinges, the stability of the structure should be preserved. The following rules should be followed:

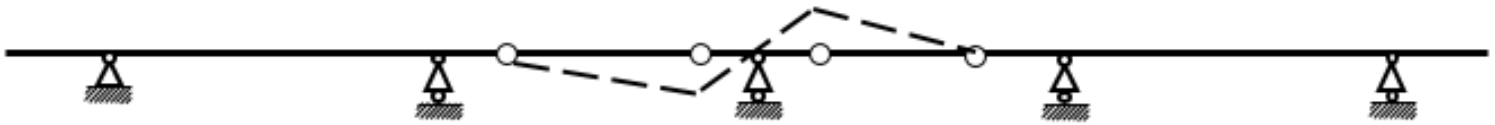
- At most two hinges should be introduced to interior spans.
- If there exists a pin support or a roller support at the end of exterior span, only one hinge should be introduced to this span.



Improper example of introducing hinges

While introducing the hinges, the stability of the structure should be preserved. The following rules should be followed:

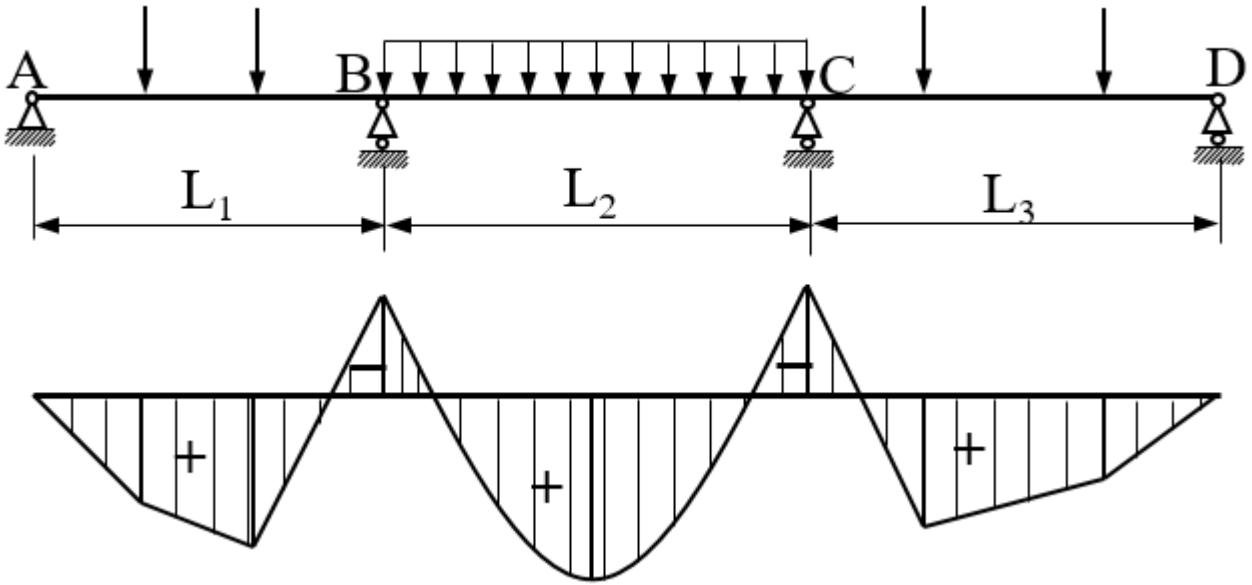
- If there exists a fixed support at the end of exterior span, at least one at most two hinges should be introduced to this span.
- If there exist two hinges at an interior span, then at most one hinge should be introduced to each neighboring span



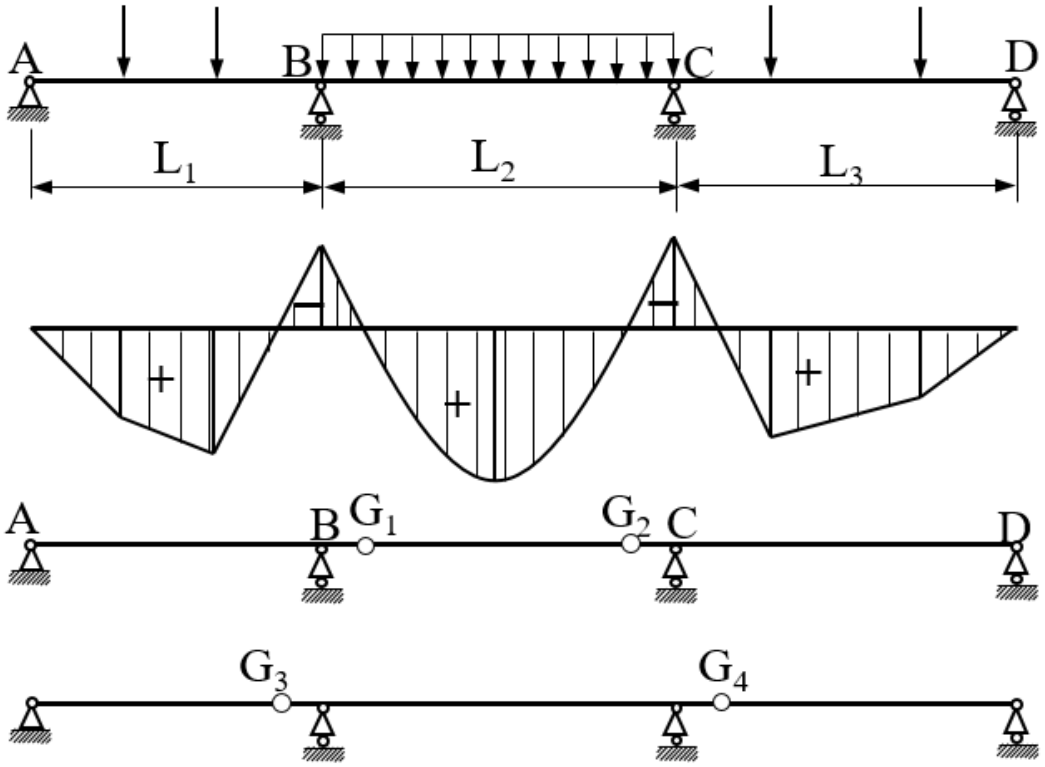
Improper example of introducing hinges

The advantages of Gerber Beams over continuous beams are as follows:

- While support settlements cause internal forces in continuous beams, they do not in Gerber beams
- Also, temperature changes do not cause any internal forces in Gerber beams
- The bending moment diagram can be modified to an desired diagram by introducing hinges at proper locations



Consider a continuous beam shown in the figure. The bending moment diagram is shown in the figure.



If Gerber beams are obtained by introducing two hinges to the sections – either in the first and third span or in the second span – at which the ending moments are zero,

the resulting bending moment diagrams will be the same as the diagram for the continuous beam. As the hinges approach to the support, the bending moment at the support decreases and the bending moment at the span increases.

Analysis of Gerber Beams

In general, there are two methods for analysis of Gerber beams:

- Solving the equilibrium equations written for the whole structure. (General Method)
- Separating the structure into parts as «carrying» and «carried» and analyzing the each part separately. (Separation Method)

1. General Method

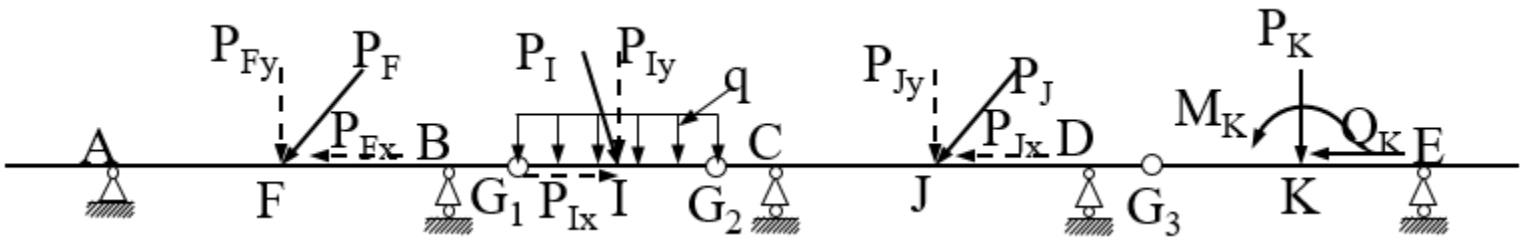
The equilibrium equations are written for the whole structure and the support reactions, then internal loads, are determined by solving these equations.

Since the introduced hinges are moment hinges, the algebraic sum of the moments of the loads on the right or left side of the hinge with respect to the hinge is zero.

In case that number of hinges is small, it is convenient to use this method. With the increasing number of equations increases and the solution will take more time.

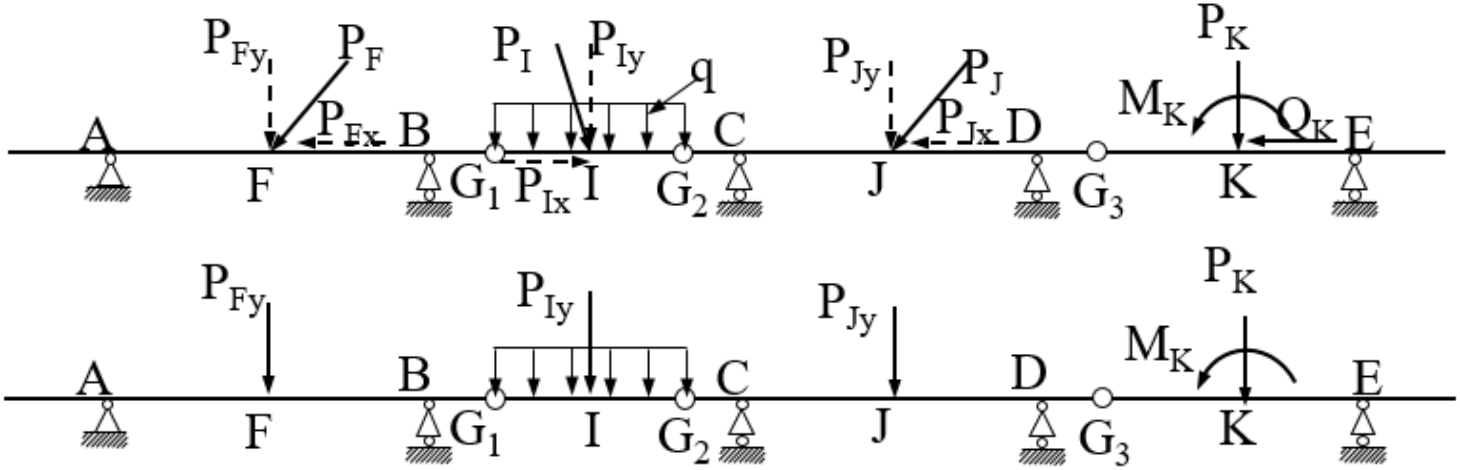
2. Separation Method

In case of structures with all of its elements on the same axis subjected to external loads acting parallel and perpendicular to the axis of element and external moments, the analysis can be carried out separately for the loads acting parallel to the axis of element and for the loads acting perpendicular to the axis of element and moments.



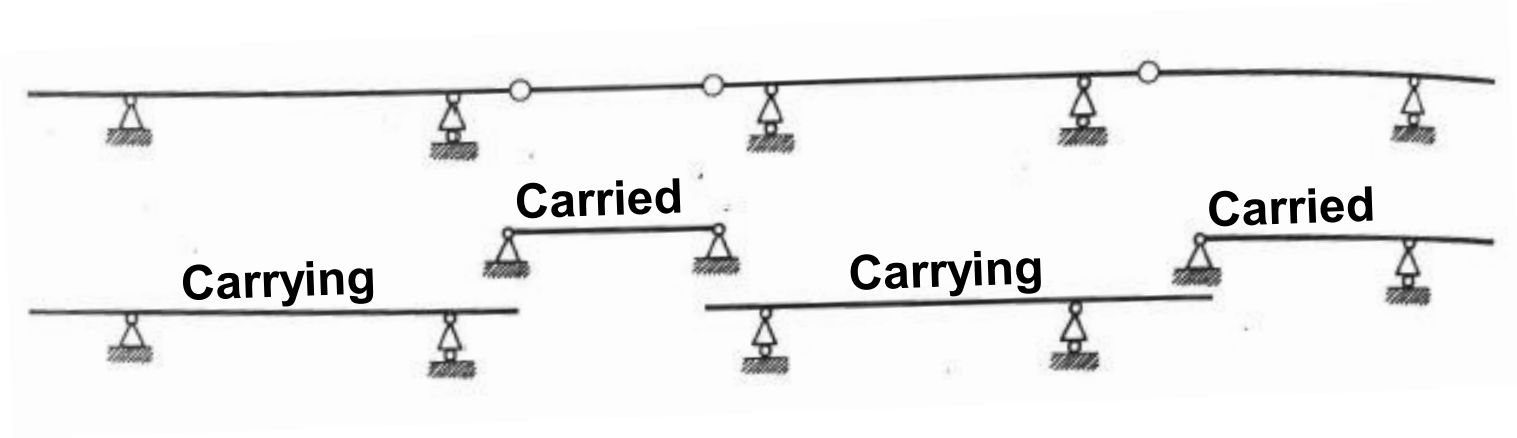
The structures subjected to loads and moment acting perpendicular to the axis of element :

Under the loads acting perpendicular to the axis of element and moment, only loads perpendicular to the axis of element exist at the Gerber beam hinges. Such Gerber beams are separated into simple structures are simply supported beams, cantilever beams and overhanging beams. A structural system layout is determined by identifying «carrying» and «carried» structures.



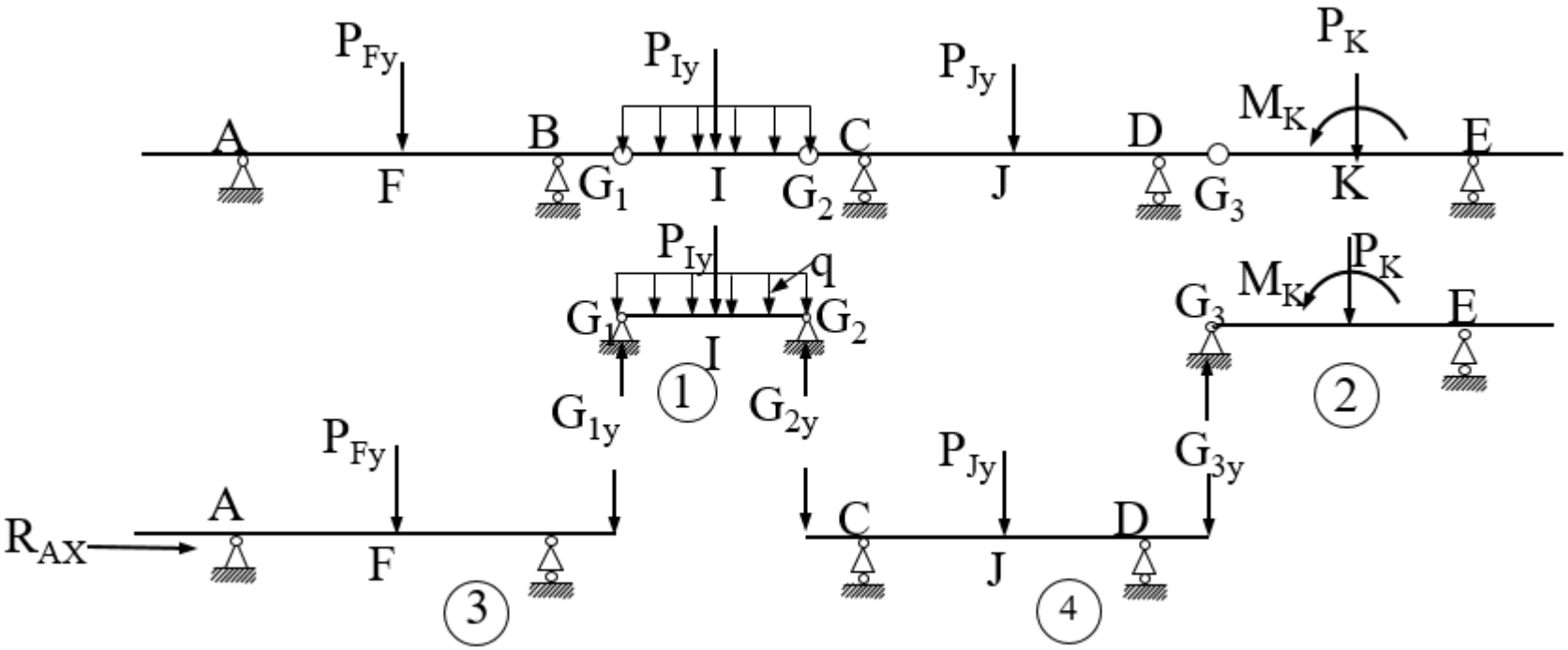
Carried structures: The structures supported by another structure such as simply supported beams or one-sided overhanging beams.

Carrying structures: The overhanging beams and the cantilever beams carrying those structures are referred to as carrying structures.



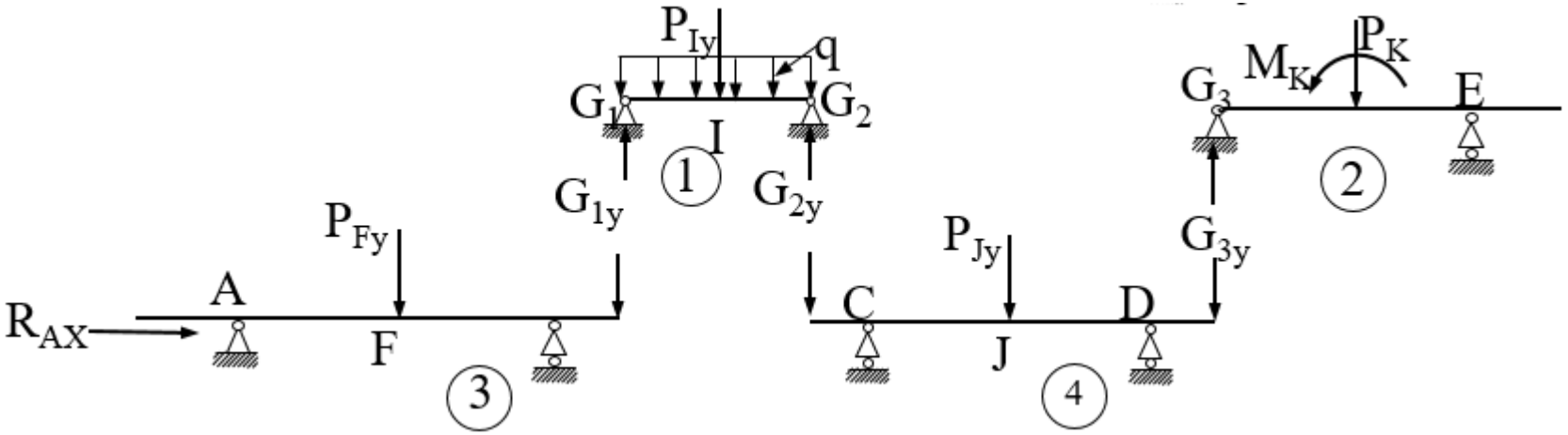
- The element between the support and the hinge in a hinged exterior span with a simple support or a roller support (G3 – E)
- The element between two hinges in a span (G1 – G2)

  are always carried structures.



- The element between the fixed support and the hinge in an exterior span (a cantilever beam)
- The element between two hinges including a span without a hinge (two – sided overhanging beam (G2 – G3)
- The element between the pin support or the roller support of an exterior span without a hinge and hinge in the neighboring interior span (A – G1)

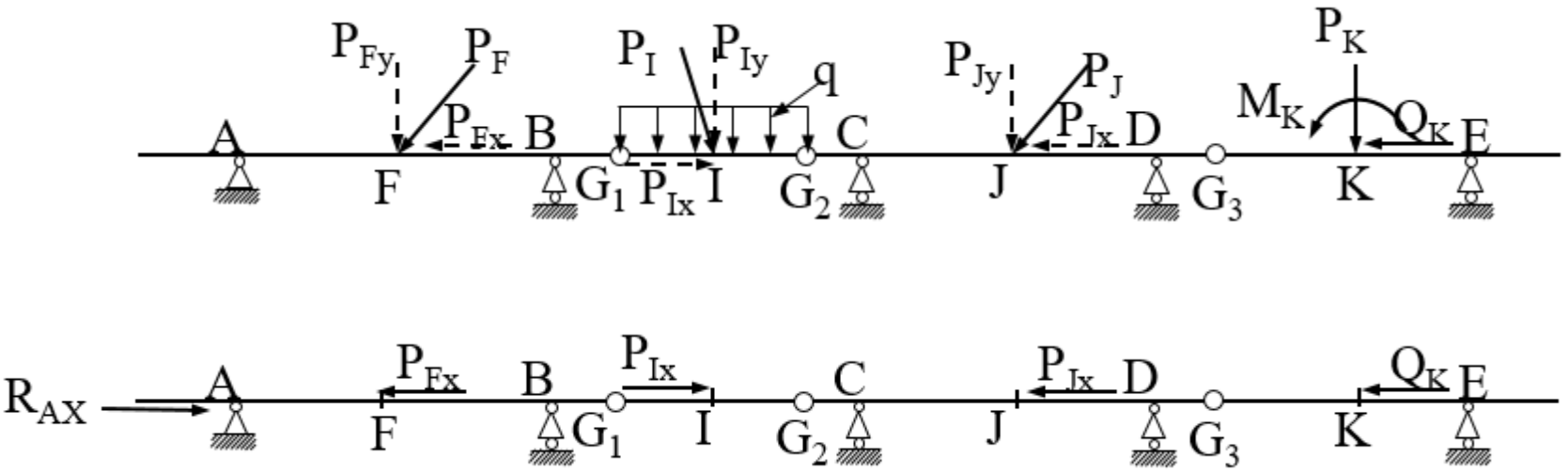
are always carrying structures.



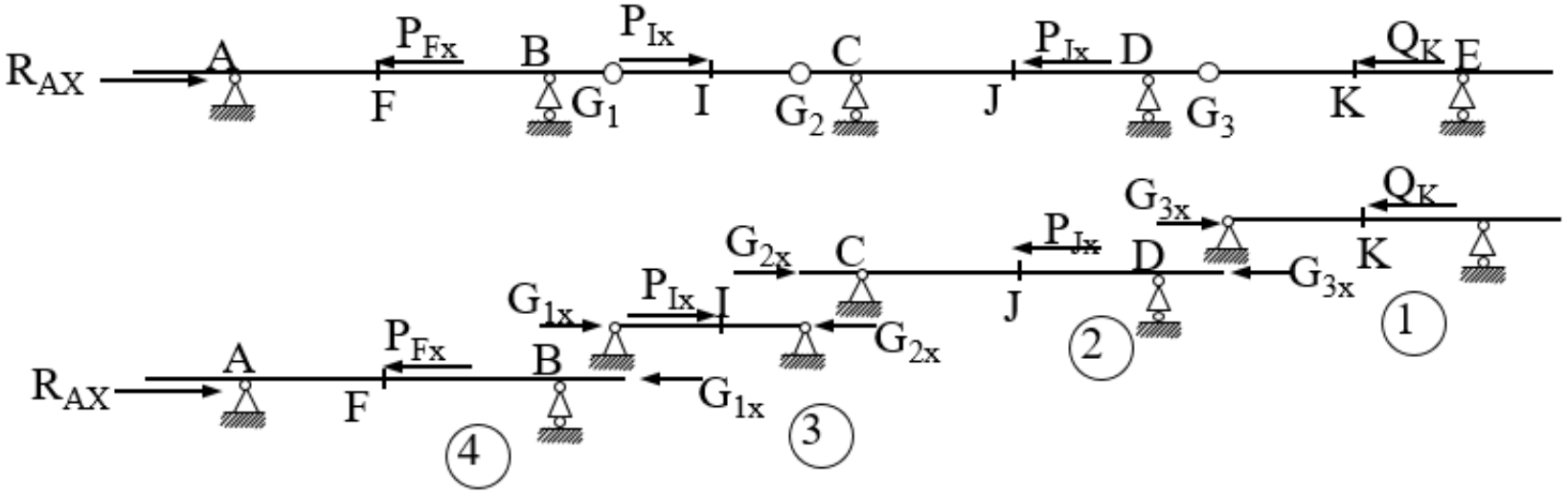
- While drawing structural system layout, «carried» structures are drawn above the «carrying» structures.
- First, carried structures are analyzed.
- Second, the hinge loads obtained from the analyses of carried structures are applied to the carrying structures as external loads and the carrying structures are analyzed.

The structures subjected to loads acting parallel to the axis of element:

The structural system layout under horizontal loads can be determined easily since the load transfer is through the fixed support or the pin support.



Loads acting parallel to the axis of element:



Analyzing the structures sequentially starting from 1, the whole structure can be analyzed.

Reference:

All the figures are taken from the book given below:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ